

Connector for ODK - User Manual

Version 1 - Administrator (includes KeSMIS Import)

Plugin version 2.0

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Homepage: <https://github.com/fnmutua/connector-for-ODK>

1. Introduction

Connector for ODK is a QGIS plugin with four tools:

Tool	Purpose
Get Data	Download ODK Central form submissions and load them as map layers
Split Layer	Split a vector layer into separate layers by attribute value
QA/QC	Run quality checks on File Geodatabase layers and export issue reports
Import	Upload QGIS vector features to a KeSMIS server

Each dialog includes a collapsible Help panel. Click << Show Help to open it.

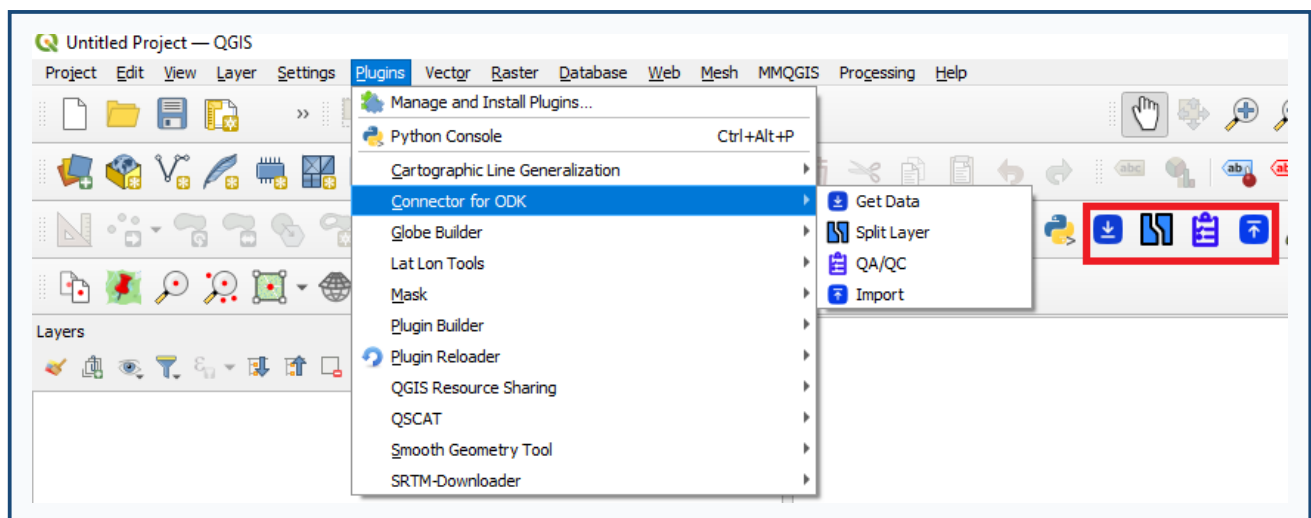


Figure 1 - QGIS menu and toolbar with Get Data, Split Layer, QA/QC, and Import

2. Prerequisites

2.1 QGIS

- QGIS 3.0 or later (desktop)
- An internet connection for ODK Central, KeSMIS, and optional package installation

2.2 Python packages

The plugin installs missing packages automatically the first time it loads. If that fails, install dependencies manually using OSGeo4W Shell from your QGIS installation (Windows).

Open OSGeo4W Shell

1. Close QGIS.
2. Open OSGeo4W Shell from your QGIS installation folder - for example C:\Program Files\QGIS 3.x\bin\OSGeo4W.bat - or from the Windows Start menu under your QGIS install (e.g. QGIS Desktop -> OSGeo4W Shell).

Use the OSGeo4W Shell that ships with the same QGIS version you use. Do not use a separate OSGeo4W install, or packages may install into the wrong Python environment.

Install required packages

Run this command in that shell:

```
python -m pip install numpy pandas geopandas fiona shapely pyproj fpdf2 requests fuzzywuzzy
rapidfuzz shortuuid openpyxl xlswriter
```

Required packages:

Package	Used for
numpy	Numerical processing
pandas	Tables and spreadsheets
geopandas	Spatial data handling
fiona	Reading/writing geospatial files
shapely	Geometry operations
pyproj	Coordinate reference systems
fpdf2	QA/QC PDF reports (import fpdf)
requests	ODK Central and KeSMIS API calls
fuzzywuzzy	Fuzzy field and attribute matching
rapidfuzz	Fast field matching (Import)
shortuuid	Unique ID generation (Import)
openpyxl	Reading dictionary.xlsx (QA/QC)
xlswriter	Writing QA/QC Excel outputs

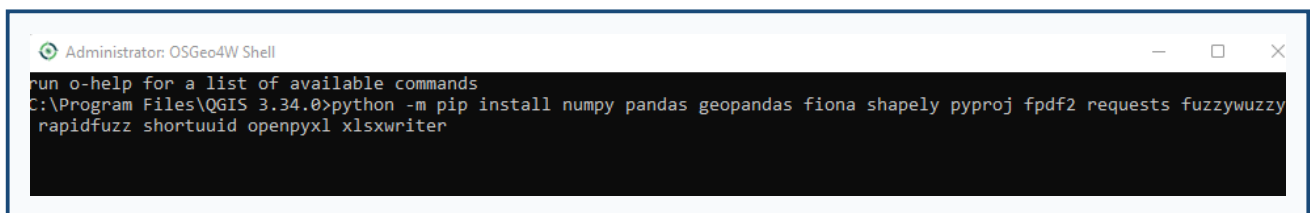


Figure 2 - Installing Python packages in OSGeo4W Shell from your QGIS installation folder

Troubleshooting: fixing a missing package error

If QGIS shows a yellow bar such as `***Couldn't load plugin 'connect_odk'***` or the Message Log reports

ModuleNotFoundError: No module named '...', a required Python package is not installed in the QGIS Python environment.

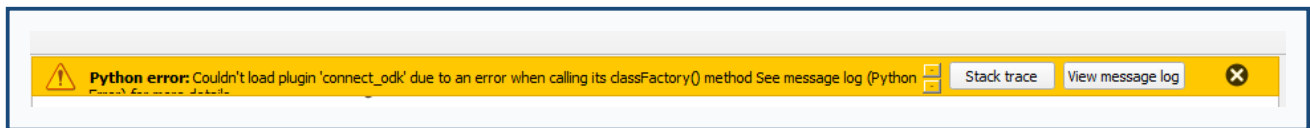


Figure 3 - Plugin load error when a required Python package is missing

Click View message log (or open View -> Panels -> Log Messages -> Python Error) to see which package is missing.

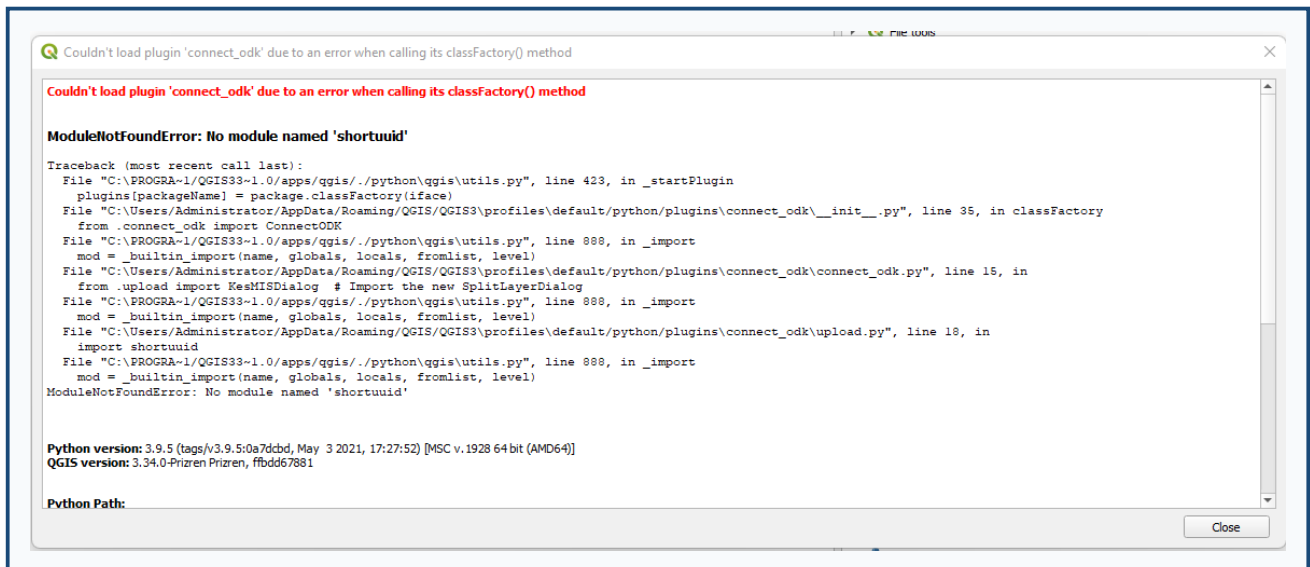


Figure 4 - Message log showing the missing package (example: shortuuid)

How to fix it

1. Note the package name in the error (the name inside quotes after No module named).
2. Close QGIS completely.
3. Open OSGeo4W Shell from your QGIS installation folder (see above).
4. Install the missing package. You can install all required packages (recommended):

```
python -m pip install numpy pandas geopandas fiona shapely pyproj fpdf2 requests fuzzywuzzy
rapidfuzz shortuuid openpyxl xlswriter
```

Or install only the missing package (replace shortuuid with the name from your error):

```
python -m pip install shortuuid
```

5. Restart QGIS. Connector for ODK should load without the error.

2.3 Data and access

Depending on which tools you use, you may also need:

- Get Data - ODK Central URL, username, and password

- Split Layer - Vector layers already loaded in the QGIS project
- QA/QC - An ESRI File Geodatabase (.gdb folder)
- Import - KeSMIS server URL, username, and password; vector layers loaded in QGIS

3. Installing the Plugin

Option A - QGIS Plugin Repository (recommended)

1. Open QGIS.
2. Go to Plugins -> Manage and Install Plugins.
3. Search for Connector for ODK.
4. Click Install Plugin.
5. Restart QGIS if prompted.

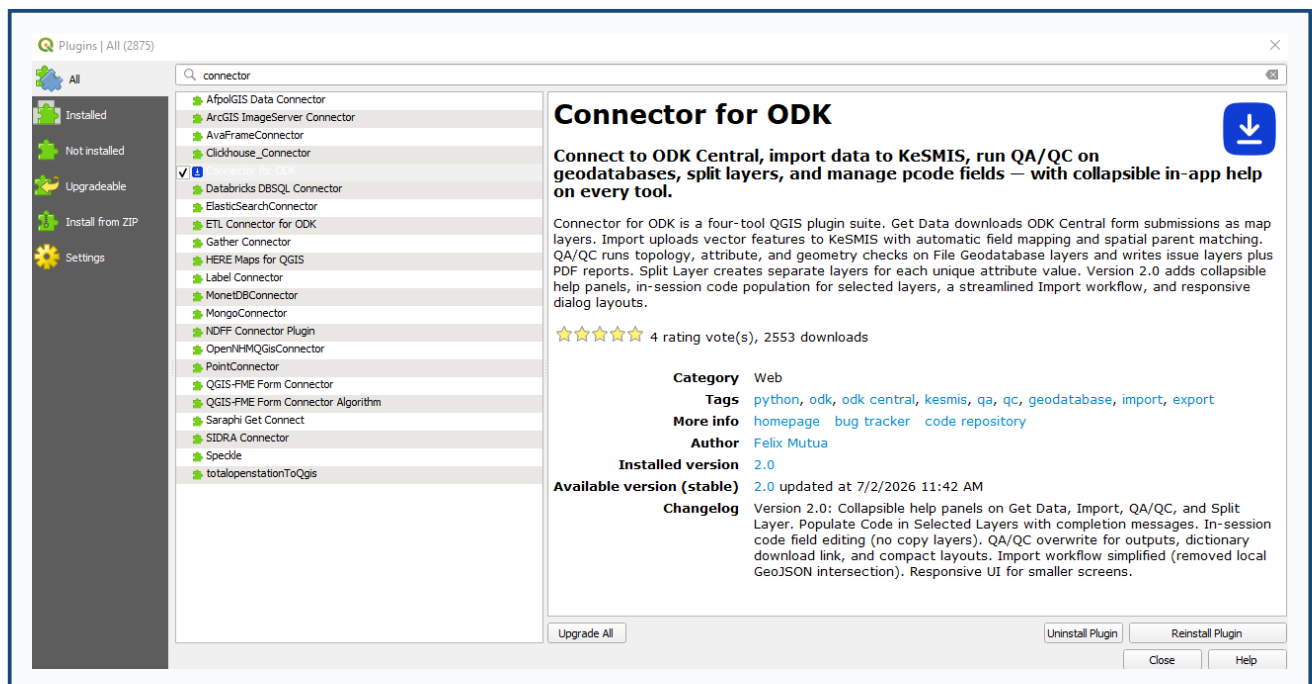


Figure 5 - Plugin Manager with Connector for ODK selected

Option B - Install from ZIP

1. Download connect_odk.zip (version 2.0).
2. In QGIS, go to Plugins -> Manage and Install Plugins.
3. Open the Install from ZIP tab.
4. Select the ZIP file and click Install Plugin.
5. Restart QGIS if prompted.

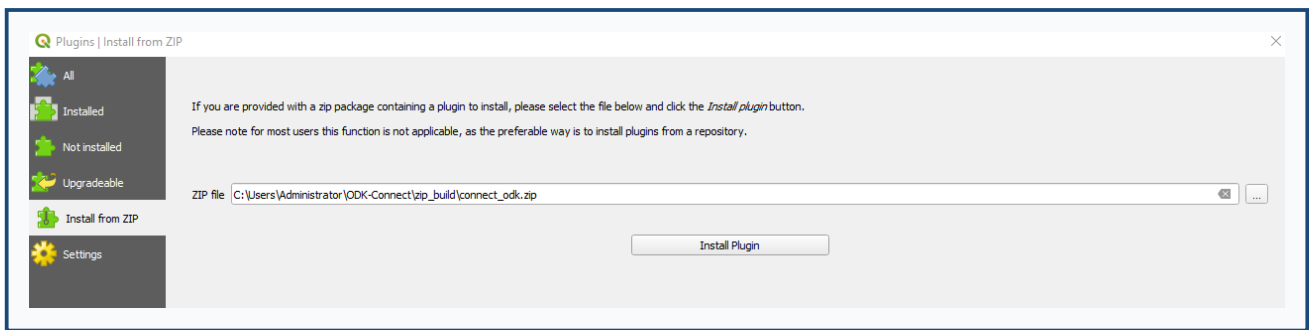


Figure 6 - Installing the plugin from a ZIP file

Verify installation

After QGIS restarts, you should see:

- Menu: Plugins -> Connector for ODK
- Toolbar icons for Get Data, Split Layer, QA/QC, and Import

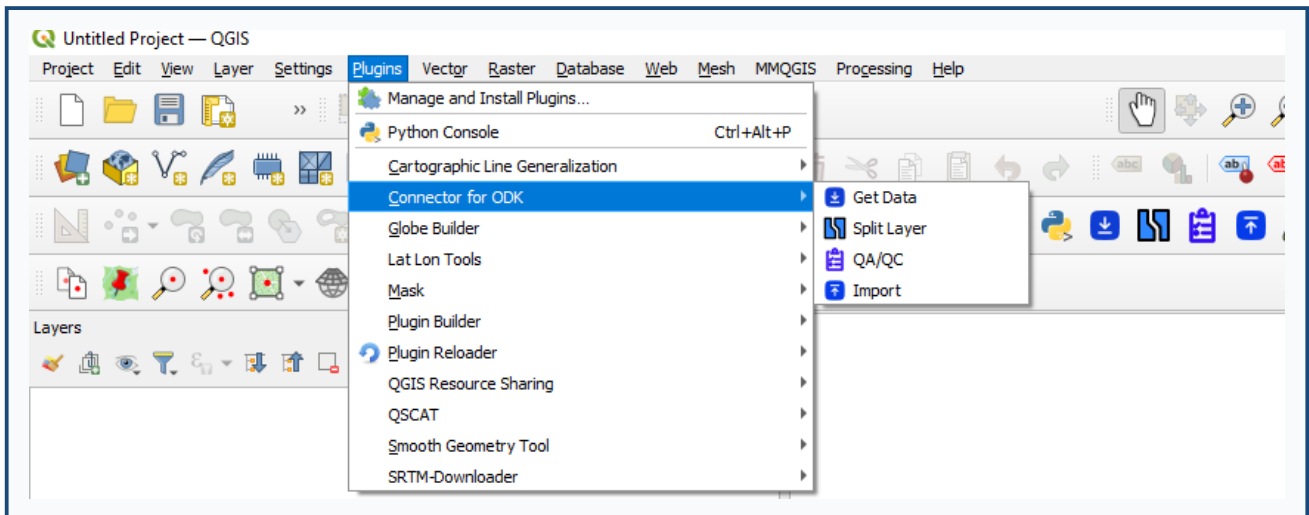


Figure 7 - Plugins menu with all four tools listed

4. Get Data (ODK Central)

Download ODK form submissions and add them to your map.

Figure 8 - Get Data dialog with ODK Central login

Steps

1. Open Plugins -> Connector for ODK -> Get Data.
2. Enter your ODK Central URL, username, and password.
3. Click Login to load projects and forms.
4. Select a project and form.
5. Click Process Form to fetch submissions and add a GeoJSON layer to the map.
6. Optionally click Get CSV to export the data as a spreadsheet.

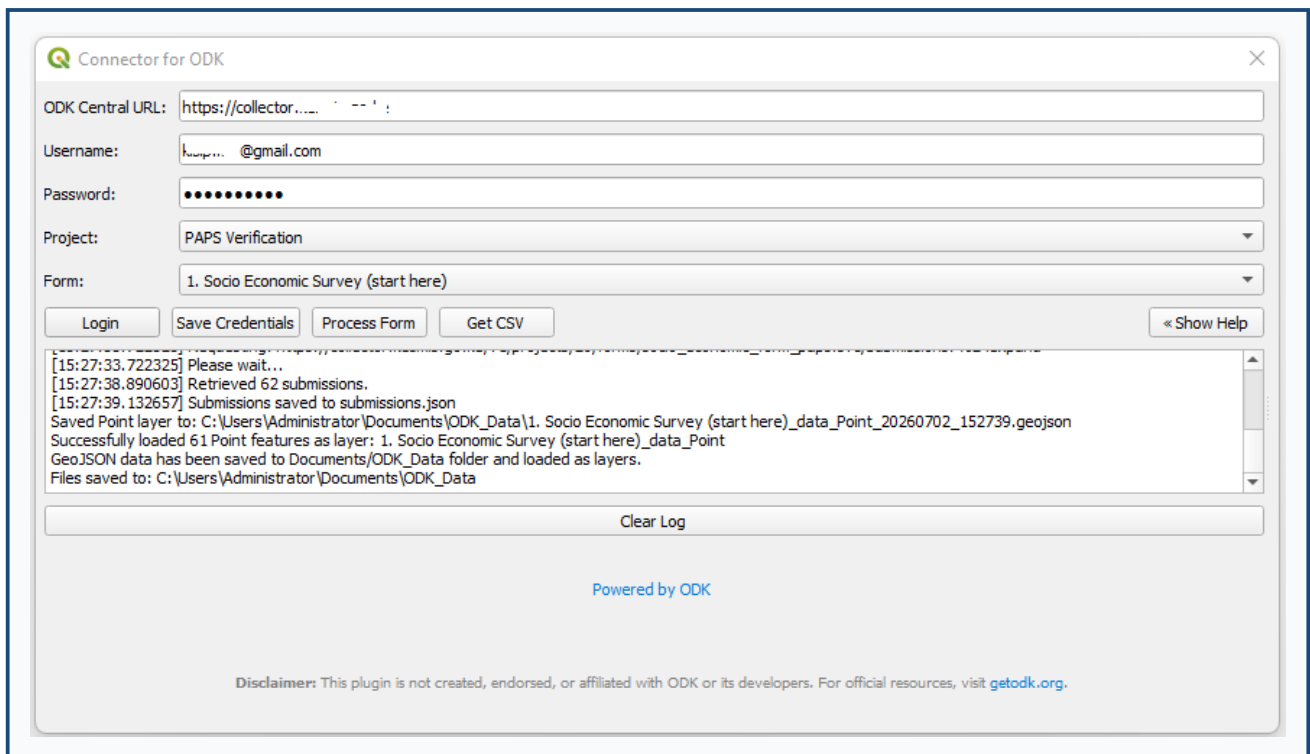


Figure 9 - Project and form selected before processing

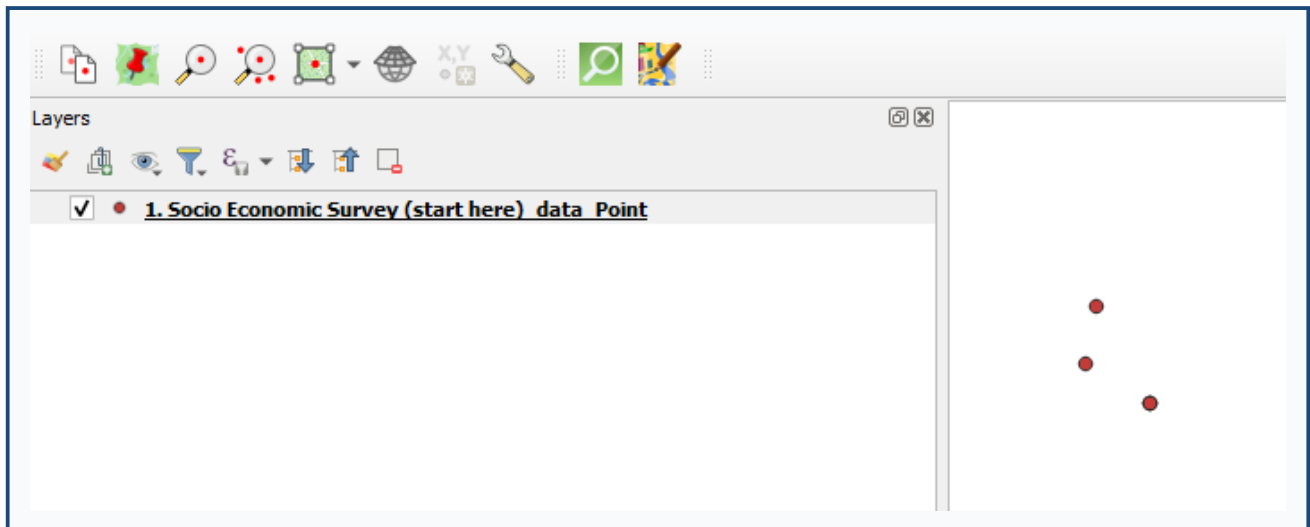


Figure 10 - Submissions loaded as a layer on the QGIS map

Notes

- Use Save Credentials to store your URL and login for next time.
- Output is in EPSG:4326 (WGS 84).
- A submissions.json file is written to the working folder.
- Check the Log panel at the bottom of the dialog for progress and errors.

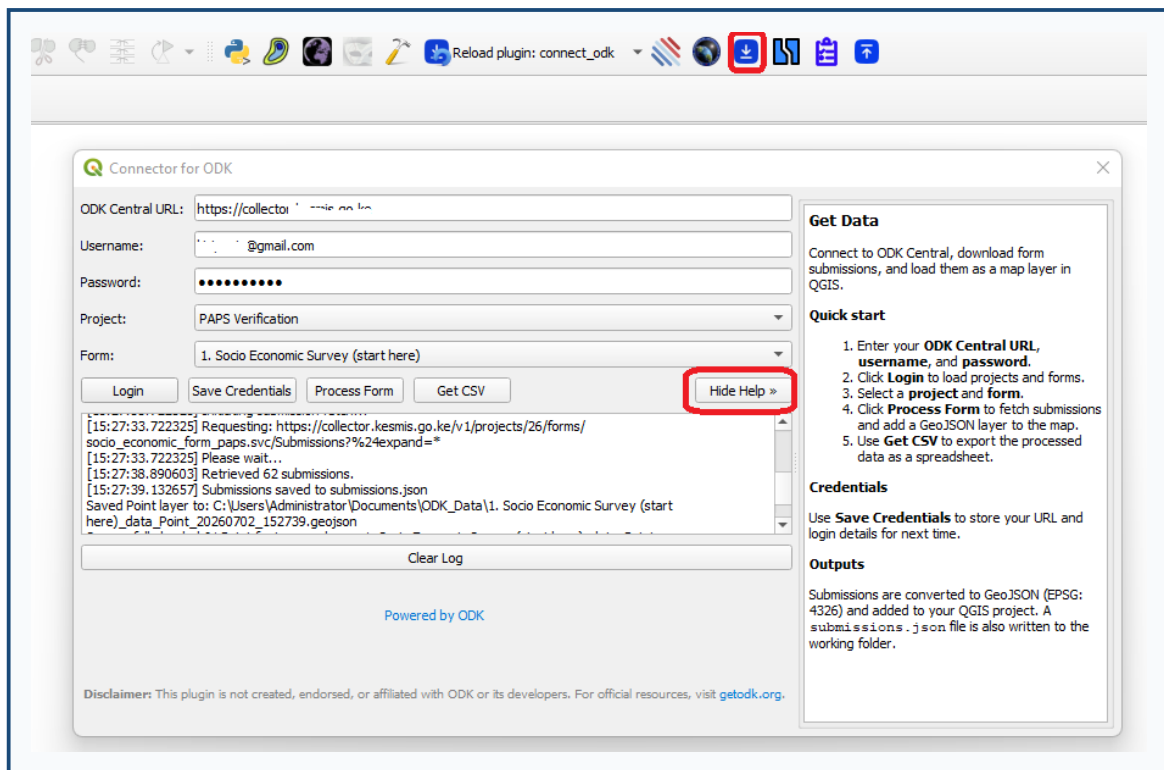


Figure 11 - Collapsible help panel in Get Data

5. Split Layer

Create separate in-memory layers for each unique value in a chosen attribute field.

Steps

1. Load the source vector layer in QGIS.
2. Open Plugins -> Connector for ODK -> Split Layer.
3. Choose a Layer from the project.
4. Choose the Field to split on.
5. Click Split Layer.

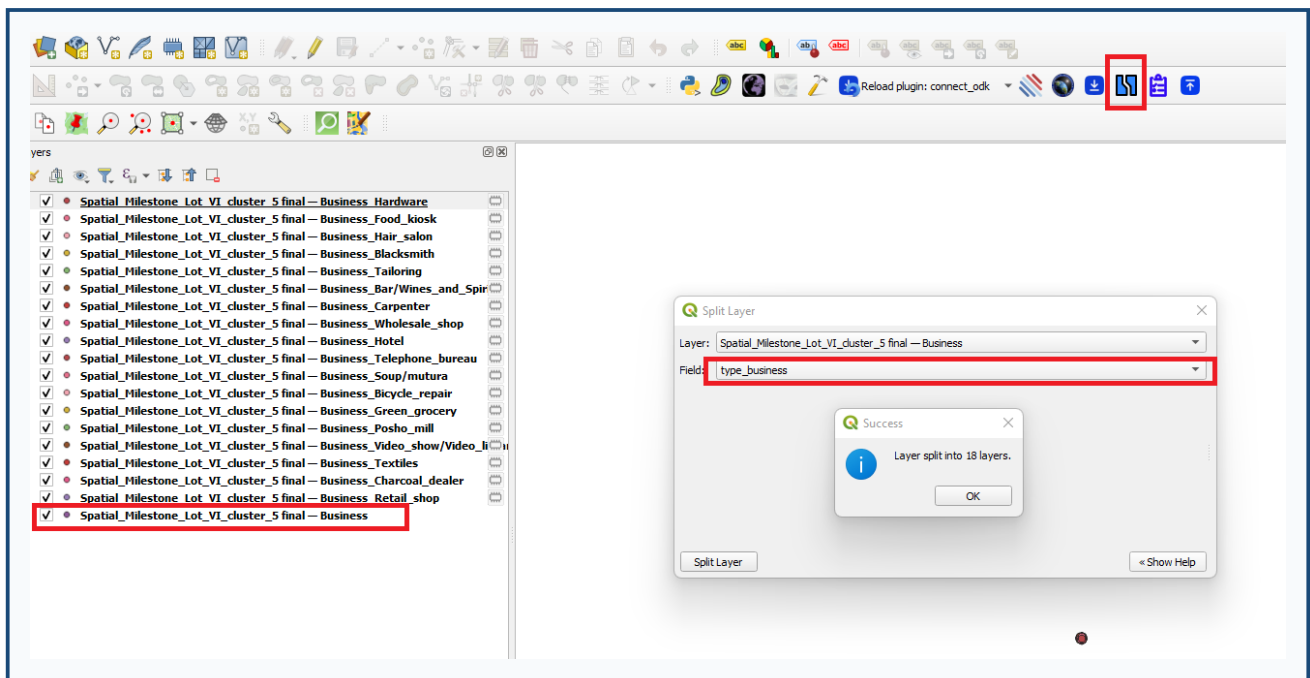


Figure 12 - Split Layer dialog

Result

After splitting:

- One new layer is created per unique non-null value.
- Layers are named {layer}_{value}.
- Empty fields are dropped from each split layer.
- Geometry and CRS are copied from the source.

6. QA/QC

Run quality checks on File Geodatabase layers and export issue layers, spreadsheets, and a PDF summary.

Steps

1. Open Plugins -> Connector for ODK -> QA/QC.
2. Click Select GeoDatabase and choose the folder containing your .gdb.
3. Click Select Output Folder for reports and issue layers.
4. Adjust parameters if needed (angle and length thresholds).
5. Under Select Layers, tick the layers to check, or use Select All.
6. Click Run All Checks.
7. When finished, use the PDF Report and Open Output Folder links below the log.

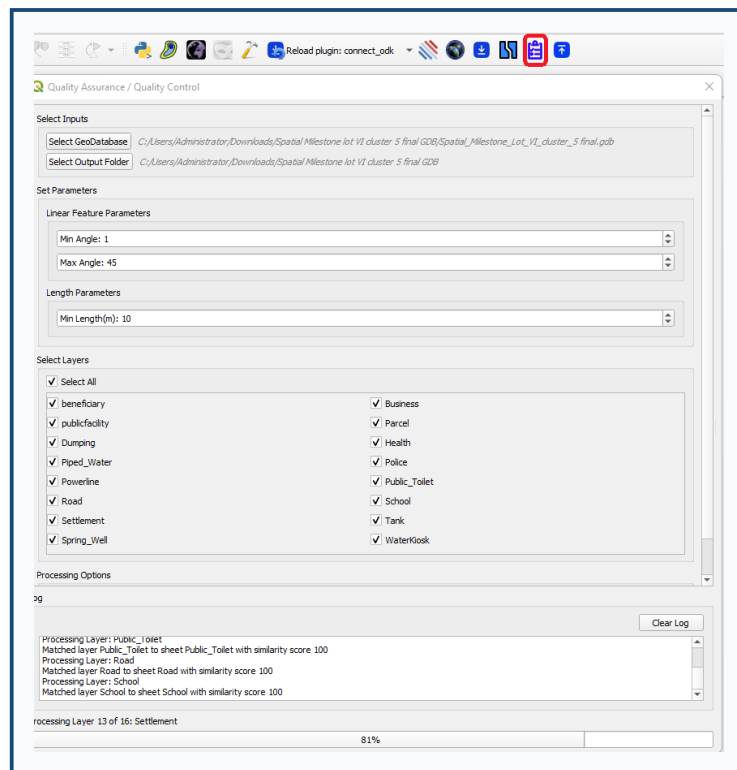


Figure 13 - QA/QC interface

Checks performed

Check	Description
Duplicate geometries	Features with identical geometry
Duplicate attributes	Rows with identical non-geometry fields
Overlapping polygons	Polygon pairs sharing area above 0.01 m2
Line issues	Sharp turns and self-intersections
Short lines	Line features shorter than the minimum length
Attribute issues	Fields validated against dictionary.xlsx

Parameters (defaults)

Parameter	Default	Purpose
Min Angle	1 deg	Lower bound for flagged turn angles
Max Angle	45 deg	Upper bound for flagged turn angles
Min Length	10 m	Flag lines shorter than this

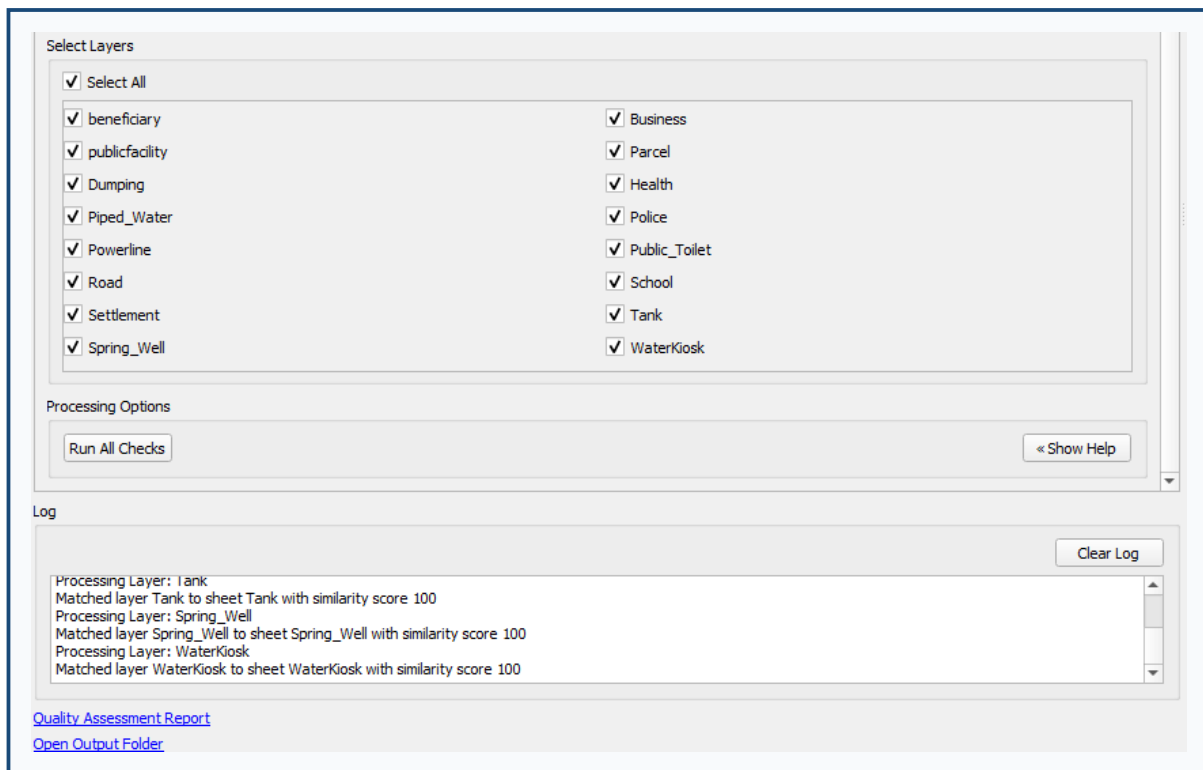


Figure 14 - Completed run with report and output folder links

Attribute dictionary

dictionary.xlsx is bundled with the plugin. Each sheet should match a layer name and include columns Attribute, Type, and optionally LEN and Options. You can download a copy from the help panel link inside the QA/QC dialog.

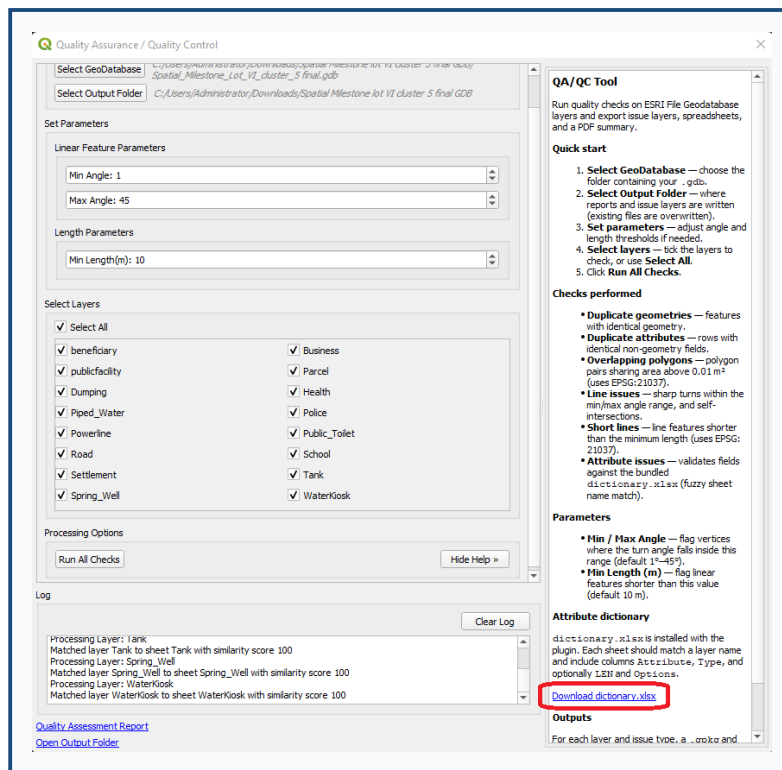


Figure 15 - Help panel showing dictionary download link

Outputs

For each layer and issue type, .gpkg and .xlsx files are written to the output folder. A summary PDF (database_summary_report.pdf) is also created. Existing output files are overwritten on re-run.

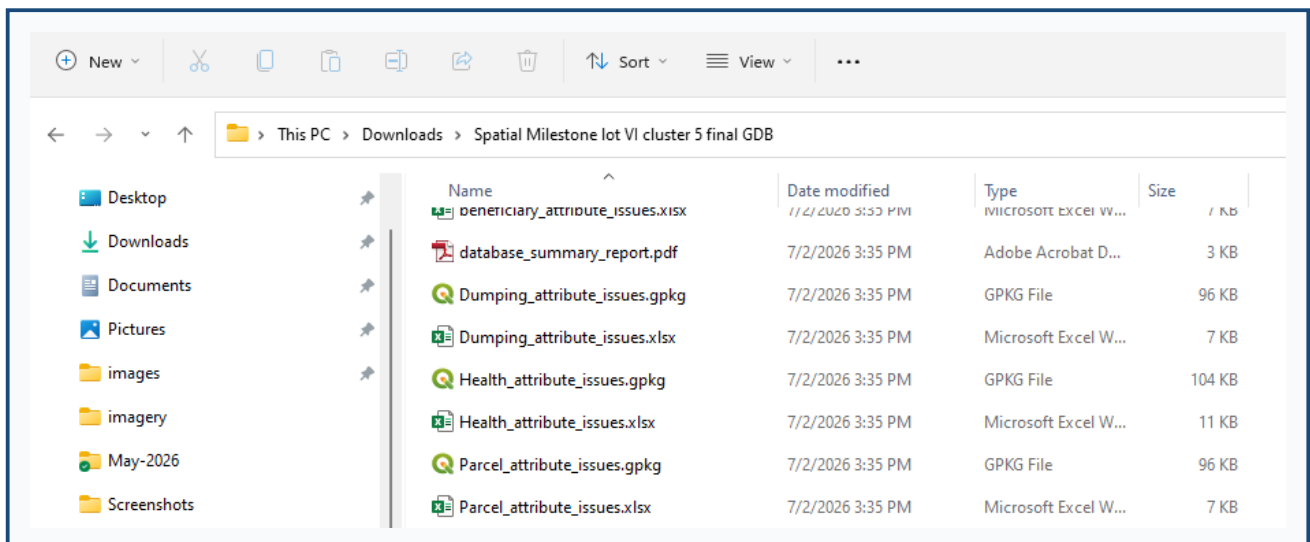


Figure 16 - Example QA/QC output files in the output folder

7. Import (KeSMIS)

Upload QGIS vector layer features to a KeSMIS server with automatic field mapping.

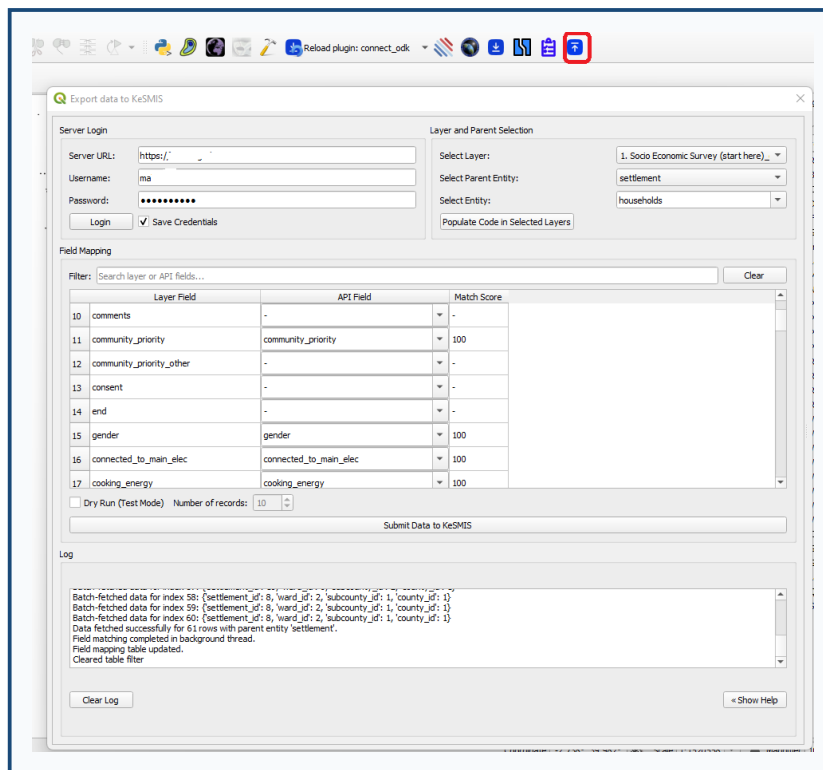


Figure 17 - Import dialog with layer, entity, and field mapping

Steps

1. Populate code in loaded layers - Load your vector layer(s) in QGIS, open Plugins -> Connector for ODK -> Import, click Populate Code in Selected Layers, tick the layers to update, and click Run. A message confirms success or reports any skipped layers (see Figure 18). Layers need a code column for pcode matching.

Export data to KeSMIS

Server Login

Server URL:

Username:

Password:

☒ Save Credentials

Layer and Parent Selection

Select Layer:

Select Parent Entity:

Select Entity:

☒ Populate Code in Selected Layers

Field Mapping

Filter:

Layer Field	API Field	Match Score
10 comments	-	-
11 community_priority	community_priority	100
12 community_priority_other	-	-
13 consent	-	-
14 end	-	-
15 gender	gender	-
16 connected_to_main_elec	conne	-
17 cooking_energy	cooking_energy	100

☐ Dry Run (Test Mode) Number of records:

Log

```

Batch-fetched data for index 58: {settlement_id: 8, ward_id: 2, subcounty_id: 1, county_id: 1}
Batch-fetched data for index 59: {settlement_id: 8, ward_id: 2, subcounty_id: 1, county_id: 1}
Batch-fetched data for index 60: {settlement_id: 8, ward_id: 2, subcounty_id: 1, county_id: 1}
Data fetched successfully for 61 rows with parent entity 'settlement'.
Field matching completed in background thread.
Field mapping table updated.
Cleared table filter
  
```

Figure 18 - Layer selection dialog for Populate Code

2. Server Login - enter the KeSMIS URL, username, and password, then click Login.
3. Select Layer - choose the QGIS vector layer to export.
4. Select Parent Entity - choose settlement or ward for spatial matching.
5. Select Entity - choose the API entity/model to submit to.
6. Review the Field Mapping table (see Figure 17). Use the filter box to search fields and adjust API field selections if needed.
7. Click Submit Data to KeSMIS.

Dry run

Enable Dry Run (Test Mode) to submit only a limited number of records before a full upload. Set the record count in the spin box next to the checkbox.

8. Tips and Troubleshooting

Issue	What to try
Plugin does not appear after install	Restart QGIS. Check Plugins -> Manage and Install Plugins -> Installed that Connector for ODK is enabled.
Package install fails	See Fixing a missing package error in Section 2.2. Close QGIS, open OSGeo4W Shell from your QGIS installation folder, install the missing package (or all packages), then restart QGIS.
ODK / KeSMIS login fails	Confirm URL, username, and password. Check network access to the server.

Import missing code field	Use Populate Code in Selected Layers before submitting.
QA/QC attribute check skipped	Ensure dictionary.xlsx has a sheet matching the layer name.
No layers in a dropdown	Load vector layers into the QGIS project first.

For updates, bug reports, and source code:

- Tracker: <https://github.com/fnmutua/connector-for-ODK>
- Repository: <https://github.com/fnmutua/ODK-Connect>